

VACA KEY, FL, USA

WMO: 997356

Lat: 24.711N

Lon: 81.107W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.7	14.3	N/A	N/A	N/A	N/A	N/A	N/A	11.6	19.6	10.5	18.7	5.3	20	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	33.2	N/A	32.4	N/A	31.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.7	230

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
9.7	8.6	7.7	DB	10.4	34.5	2.1	1.3	8.9	35.5	7.6	36.3	6.4	37.0	4.9	38.0	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.7	21.1	22.3	23.3	25.1	26.9	28.7	29.3	29.5	28.8	27.1	23.8	22.5
	DBStd	3.65	3.19	3.16	2.42	1.89	1.31	1.29	1.21	1.12	1.24	1.88	2.56	3.26
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	33	14	8	2	0	0	0	0	0	0	0	1	8
	CDD10.0	5732	343	344	412	453	523	561	598	604	563	531	414	386
	CDD18.3	2724	98	119	156	203	264	311	340	346	313	272	164	137
	CDH23.3	26669	301	494	675	1417	2594	3822	4462	4622	3898	2817	949	618
	CDH26.7	8907	8	18	49	181	533	1501	2018	2177	1552	737	99	35
Wind	WSAvg	3.3	4.0	3.2	3.5	3.2	3.1	2.5	2.3	2.3	2.8	4.0	4.3	3.7
Precipitation	PrecAvg	1036	50	43	39	50	73	117	93	150	179	150	45	46
	PrecMax	1406	209	154	90	156	231	373	212	491	385	437	169	137
	PrecMin	706	0	1	8	2	12	6	21	27	92	39	0	1
	PrecStd	187	51	40	25	38	61	85	47	85	62	105	46	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.5	28.0	29.4	31.5	31.6	33.3	34.0	34.1	33.6	32.4	29.7	28.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	26.6	27.2	28.0	29.4	30.4	32.1	33.0	33.0	32.4	30.8	28.7	27.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	25.8	26.6	26.9	28.3	29.6	31.4	32.1	32.1	31.5	30.1	27.9	26.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	4.2	4.0	3.8	3.8	3.6	4.0	4.1	4.4	4.2	3.4	3.3	3.6
		MCDBR	4.7	4.0	4.6	4.6	4.8	5.0	5.1	5.2	5.3	4.4	3.8	3.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Clear-Sky Solar Irradiance	taub	0.363	0.377	0.383	0.401	0.425	0.443	0.451	0.434	0.423	0.409	0.378	0.370
taud		2.549	2.509	2.478	2.406	2.392	2.410	2.388	2.460	2.457	2.486	2.518	2.547	
Ebn at Noon		897	906	916	900	872	849	842	860	865	863	872	873	
Edh at Noon		92	102	110	121	122	119	121	113	111	103	94	89	
All-Sky Solar Radiation	RadAvg	3.90	4.82	5.92	6.68	6.73	6.24	6.37	6.06	5.37	4.82	4.17	3.67	
	RadStd	0.29	0.32	0.24	0.30	0.59	0.48	0.30	0.24	0.25	0.33	0.21	0.31	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (21 neighbors)	+0.38	+0.87	N/A	N/A	N/A	N/A	N/A	-13	+116	+104

Nomenclature: See separate page